

Zero-counting lower bounds and measured K - ϵ curves for approximate Gaussian rank

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Working note for the theory companion to the preprint doi:10.5281/zenodo.21457048. Vocabulary and gate history: issue #71.

1. Introduction

How well can a continuous-variable quantum state be described by finitely many Gaussians? The exact and approximate *Gaussian rank* of a pure state — the minimum number of pure-Gaussian components in a superposition — has recently been defined and developed by Hahn, Takagi, Ferrini and Yamasaki [HTFY, arXiv:2404.07115] (χ and its approximate version χ_δ , Gaussian extent ξ), with the fidelity-ball ϵ -relaxation template established for the stellar rank by Hahn, Garnier, Ferrini, Ferraro and Chabaud [arXiv:2410.23721] and lower-bound techniques for the *coherent-state* rank initiated by Cottier and Chabaud [arXiv:2604.00766]. **We claim no novelty for the definitions;** this note adopts and combines them (pure χ from HTFY; the fidelity-ball and sup-roof forms from the stellar-rank line) and contributes three things the cited line does not currently contain:

1. **Certified lower bounds from Bargmann zeros for bounded common-squeezing dictionaries** (§5), by a technique different from the low-rank-approximation route of [Cottier–Chabaud]: robust zeros of the target’s Bargmann function survive fidelity-ball perturbation (Rouché) and are counted against an exponential-sum zero budget. The restriction to a bounded dictionary is **forced, not cosmetic**: we exhibit counterexample families showing that no disk-zero argument can bound the unrestricted rank (§5.4).
2. **An inequivalence theorem** (§6): the thermal lossy cat state of the preprint’s experiment 20 has approximate Gaussian rank ≤ 2 for every ϵ , yet *no finite-rank pre-loss operator representation at any efficiency* (exact statement below). Gaussian-superposition compressibility and pre-loss operator-rank compressibility are inequivalent notions, and the same state separates them.
3. **Measured K - ϵ -type curves on public data and synthetic gates** (§7): the cutoff-scored family-constrained approximation curve and the real-data rank proxy of experiment 22, and (Gate E, experiment 23) a robust-zero census for finite-energy GKP states (numerical status per §7.2).

Throughout, three epistemic labels are used and never mixed: **certified** (theorem-backed), **empirical** (best-found by optimization; an upper bound on what is achievable), **proxy** (measured quantity standing in for an inaccessible one, stated with its model assumptions).

2. Definitions

2.1 Gaussian rank and its fidelity-ball relaxation (adopted)

Fix a finite mode number M . For a normalized pure state $|\psi\rangle$,

$$\chi_G(|\psi\rangle) := \min\{K : |\psi\rangle \propto \sum_{k=1}^K c_k |G_k\rangle, |G_k\rangle \text{ pure Gaussian}\},$$

with $\chi_G(|\psi\rangle) := \infty$ when no finite decomposition exists; the value set is $\mathbb{N}_{\geq 1} \cup \{\infty\}$ [HTFY’s χ]. For a density operator ρ , using generalized ensembles (a measure μ on a standard Borel space with a measurable field of pure states, $\rho = \int |\psi_i\rangle\langle\psi_i| d\mu(i)$):

$$\chi_G(\rho) := \inf \text{ over decompositions of } [\text{ess sup}_i \chi_G(|\psi_i\rangle)],$$

the essential supremum taken with respect to the completed measure. (This sup-type roof follows the *stellar-rank* convention of [arXiv:2410.23721]; HTFY’s mixed-state treatment is extent-based, and — importantly — HTFY **do** define continuous-decomposition versions on the extent side (their Eq. 82 ξ' and Eq. 85–86 Ξ vs Ξ'), explicitly leaving “the analysis of the difference in these definitions for future work”. The mixed χ_G used here is thus not more permissive than anything in the cited literature as a *concept*; what is ours is the measure-theoretic ess-sup **rank** roof formulation together with its attainment (Lemma A1') and channel monotonicity (Theorem A') — a partial answer, on the rank side, to the discrete-vs-continuous question HTFY left open. The coherent-POVM proof of A' needs the continuous form; whether the discrete and continuous rank roofs coincide is open, cf. issue #122.) Because the objective is integer-valued, the outer infimum is attained (Lemma A1'). Convex mixing is free by design; this matches the classical-simulability anchor (mixtures of Gaussians are the classical backdrop against which components are counted).

With F the **squared Uhlmann fidelity** $F(\rho, \sigma) = (\text{Tr} \sqrt{(\sqrt{\rho} \sigma \sqrt{\rho})})^2$ (pure-pure case $|\langle \psi | \phi \rangle|^2$), the approximate rank is

$$K_\varepsilon^{\wedge}\{G, F\}(\rho) := \min\{\chi_G(\sigma) : F(\rho, \sigma) \geq 1 - \varepsilon\}, \varepsilon \in [0, 1],$$

monotone non-increasing in ε , $K_0 = \chi_G$. This is the fidelity-ball template of [arXiv:2410.23721, Def. 7] applied to χ_G ; on pure states it is interconvertible with χ_δ of [HTFY] via $\|\psi - \psi'\|^2 = 2(1 - \text{Re}\langle \psi | \psi' \rangle)$ (Appendix C).

2.2 The bounded common-squeezing dictionary (the certified results' quantity)

Single mode. Sections 4–5 and the rows D2–D3 are single-mode statements (one-variable Bargmann functions, zeros in a complex disk); the multimode extension is open (§8).

The certified lower bounds of §5 concern the following *restricted atomic dictionary*. Fix one complex squeezing parameter a , $|a| \leq a^* < 1$ — the same a in **modulus and angle** for every atom — and a displacement budget B . The **atoms** are single normalized Gaussian kets:

$$\mathcal{A}(a, B) := \{ |G\rangle : \text{Bargmann function } F_G(z) \propto e^{\{a z^2/2 + b z\}}, |b| \leq B \}.$$

The **rank** counts atoms: $\chi_{G_{\text{eq}}(a, B)}(|\psi\rangle) := \min\{K : |\psi\rangle \propto \sum_{k=1}^K c_k |G_k\rangle, |G_k\rangle \in \mathcal{A}(a, B)\}$ (values $\mathbb{N}_{\geq 1} \cup \{\infty\}$); the mixed case is the restricted ess-sup roof over generalized ensembles whose members admit such decompositions **with the one fixed a of the dictionary instance** (members may not vary a ; a union-over- a variant would also satisfy Theorem B', whose proof is uniform in a , but is not the quantity used here). $K_\varepsilon^{\wedge}\{G_{\text{eq}}(a, B), F\}$ is then defined verbatim as in §2.1. **Physical translation** (operator ordering $D(\alpha)S(\xi)|0\rangle$, $\xi = r e^{i\varphi}$, $S(\xi) = \exp[(\xi^* \hat{a}^2 - \xi \hat{a}^\dagger{}^2)/2]$): $a = -e^{i\varphi} \tanh r$ and $b = \alpha - a \alpha$, *an invertible real-linear map at fixed a with $(1-|a|)/|a| \leq |b| \leq (1+|a|)/|a|$ and $\alpha = (b + a b)/(1-|a|^2)$* . So at fixed a , a displacement bound and a B -bound are equivalent, and with $\langle n \rangle = |a|^2 + \sinh^2 r$ the dictionary is energy-bounded; **neither equivalence is uniform as $|a| \rightarrow 1$** , which is why $a^* < 1$ is part of the definition.

Since $K_\varepsilon^{\wedge}\{G, F\} \leq K_\varepsilon^{\wedge}\{G_{\text{eq}}(a, B), F\}$, lower bounds on the restricted quantity do **not** transfer to the unrestricted one. §5.4 shows this is intrinsic to count-only zero arguments, not a defect of this particular proof.

2.3 The subordinate contrast quantity R_ε

For the inequivalence theorem only, define the **pre-loss operator rank**

$$R_\varepsilon(\rho) := \min\{R : \exists \eta' \in (0, 1], \exists \rho' \geq 0 \text{ with } \text{Tr } \rho' = 1, \text{rank}(\rho') \leq R \text{ and } F(\text{loss}_{\{\eta'\}}(\rho'), \rho) \geq 1 - \varepsilon, \\ \text{with } \min \emptyset := \infty\}.$$

(The normalization keeps F inside its density-operator domain — without it the value could be manipulated by scalar rescaling — and the ∞ convention is what makes statements like $R_0 = \infty$ well-formed.)

R_ε is a repository-specific contrast quantity with no counterpart in the cited literature. **It is subordinate by charter:** the note’s claims are made for χ_G / K_ε ; R_ε appears only to exhibit their inequivalence. It is trivially

non-increasing in ε ; no **Gaussian-channel** monotonicity is claimed for it (loss cannot be “pulled back” through the monotonicity of §3; that inequality only runs forward).

2.4 Declaration table (single source of truth for “which quantity, which state”)

Every theorem, table and figure in this note cites one row of this table. Rows are also targets of the build-time claim checker (§7.3): the coded checks compare numbers that prose sentences attribute to rows against the artifact JSONs they name.

ID	Quantity	Target state	Modes	Dictionary / family	Source of numbers
D1	$K_\varepsilon^{\{G,F\}}$ (unrestricted)	any ρ	any finite M (definitions only)	all pure-Gaussian superpositions, roof	definitions only; no disk-zero lower bound is claimed (§5.4a rules out that technique, not other routes — cf. Cottier–Chabaud) Theorem B' + exp23
D2	$K_\varepsilon^{\{G_{eq}(a,B),F\}}$ (restricted)	pure targets of §5; ψ_{trunc} of Gate E	single mode	common complex a ($ a \leq a^* < 1$), $ b_k \leq B$	$N_{robust_bounded}$
D3	$K_\varepsilon^{\{G_{eq}(a,B),F\}}$ lifted to the untruncated finite-energy comb	untruncated finite-energy GKP comb	single mode	as D2, threshold $d(\varepsilon) + \ \text{Fock tail}\ + 4 \cdot \ \text{lattice remainder}\ $ (normalization factor, §7.2)	exp23 $N_{robust_lifted_bounded}$ (computed column; per-configuration)
D4	R_ε (pre-loss operator rank; subordinate)	thermal lossy cat ρ_T	single mode	loss $\circ$ finite-rank	exp20 theorems ($R_0 = \infty$, certified). No empirical R_ε point is claimed: the Route B residuals are cutoff-conditioned generalized fidelities, which can exceed the true infinite-dimensional fidelity, so they live in row D5
D5	family- constrained approximation curve	thermal lossy cat ρ_T	single mode	rank-2 BB† + loss, K kets/column	exp22 JSON (empirical; cutoff-conditioned)
D6	held-out NLL rank proxy	measured GKP data (true state unknown)	single mode	operator mixture rank R, K=4 kets	exp22 JSON (proxy)

3. Consistency and monotonicity (certified)

Lemma A1'. χ_G is well-defined with values in $\mathbb{N}_{\geq 1} \cup \{\infty\}$; the sets $\{\psi : \chi_G(\psi) \leq K\}$ are universally measurable; for every ρ the outer infimum in the roof is attained; $K_\varepsilon^{\{G,F\}}$ is non-increasing in ε with $K_0^{\{G,F\}} = \chi_G$.

Proofsketch (full version in Appendix A). Measurability: $\{\chi_G \leq K\}$ is the projection of a Borel set in the $(K\text{-tuple of Gaussian parameters} \times \text{coefficients})$ parameterization, hence analytic, hence universally measurable; the ess sup is taken in the completed measure. Attainment: the roof objective takes values in a discrete set; if no decomposition achieved the value $m := \inf$, every decomposition would have value $\geq m+1$, contradicting $\inf = m$. $K_0 = \chi_G$ follows from $F(\rho, \sigma) = 1 \Leftrightarrow \rho = \sigma$. ■

Theorem A' (Gaussian-channel monotonicity; row D1). For every Gaussian channel Λ (including loss and classical Gaussian noise), every state ρ , and every $\varepsilon \in [0,1]$:

$$K_{\varepsilon}^{\Lambda}\{G,F\}(\Lambda\rho) \leq K_{\varepsilon}^{\Lambda}\{G,F\}(\rho), \text{ and } \chi_G(\Lambda\rho) \leq \chi_G(\rho).$$

Proof. (i) Rank part. If $\chi_G(\sigma) = \infty$ there is nothing to prove; let σ achieve $\chi_G(\sigma) = m < \infty$ with decomposition $\{\mu, \psi_i\}$, $\text{ess sup } \chi_G(\psi_i) \leq m$ (attained by Lemma A1'). Stinespring-dilate $\Lambda = \text{Tr}_E [U_G (\cdot \otimes |0\rangle\langle 0|_E) U_G^\dagger]$ with U_G Gaussian (for a mixed Gaussian environment, purify with a further Gaussian ancilla). Resolve the environment in the coherent-state POVM: for each i and $\beta \in \mathbb{C}^{\{M_E\}}$,

$$M_\beta \psi_i := \langle \beta |_E U_G (|\psi_i\rangle \otimes |0\rangle_E)$$

is (unnormalized) a superposition of at most $\chi_G(\psi_i) \leq m$ pure Gaussian kets, because U_G maps Gaussian products to Gaussians and $\langle \beta |_E$ projections of Gaussians are Gaussian. The joint measure $d\nu(i, \beta) = d\mu(i) p(d\beta|i)$, with zero-norm branches discarded and states renormalized (the weight absorbed into ν), is a generalized ensemble for $\Lambda\sigma$. Hence $\chi_G(\Lambda\sigma) \leq m$. (ii) ε part. If $F(\rho, \sigma) \geq 1-\varepsilon$ then $F(\Lambda\rho, \Lambda\sigma) \geq F(\rho, \sigma) \geq 1-\varepsilon$ by CPTP monotonicity of the Uhlmann fidelity; combine with (i). ■

The same proof restricted to $G_{\text{eq}}(a,B)$ requires the channel to preserve the dictionary and is **not** claimed; monotonicity is a statement about the unrestricted quantities.

4. Zero counting for the dictionary (certified)

Lemma P1 (multiplicity). A nonzero exponential sum $g(z) = \sum_{k=1}^K c_k e^{b_k z}$ with distinct b_k and $c_k \neq 0$ has zeros of multiplicity at most $K-1$.

Proof. $g^{(j)}(z_0) = \sum_k c_k b_k^j e^{b_k z_0}$; if all derivatives up to order $K-1$ vanish, the Vandermonde matrix (b_k^j) annihilates the nonzero vector $(c_k e^{b_k z_0})$ — contradiction. ■

Lemma B1'-eq (disk zero count). Let $f \not\equiv 0$ have the dictionary form $f(z) = e^{az^2/2} \sum_{k=1}^K c_k e^{b_k z}$ with $|b_k| \leq B$ (repeated exponents merged, so the reduced order is $\leq K$). Then for every disk $\bar{D}(z_0, R)$, counting multiplicity:

$$N(\bar{D}(z_0, R); f) \leq C \cdot (K + B \cdot R), \text{ with } C \text{ an absolute constant.}$$

Proof. $e^{az^2/2}$ is zero-free, so $N(f) = N(g)$ for the exponential sum g . Apply the disk zero bound for exponential polynomials [O. Friedland, arXiv:2606.24823, Lemma B.1]: a nonzero exponential sum of order m with $\sigma = \max_k |\mu_k|$ has at most $C(m + \sigma r)$ zeros (with multiplicity) in any closed disk of radius r . Here $m \leq K$, $\sigma \leq B$. ■

Because [arXiv:2606.24823] is a recent preprint, Appendix B re-derives its Lemma 3.2 (disk growth: $\sup_{\{D(z_0, R)\}} |q| \leq e^{(2R\sigma)(CR/r)} \{m-1\} \sup_{\{D(z_0, r)\}} |q|$, by reducing rays $z = z_0 + s\theta$ to the classical interval Turán inequality — the exponents $\theta\mu_k$ satisfy $|\text{Re}(\theta\mu_k)| \leq \sigma$) and its Lemma B.1 (Blaschke: N zeros in $D(z_0, r)$ force a factor θ^N , $\theta < 1$ absolute, against the disk growth at outer radius $8r$). The only external input then remaining is the classical interval Turán inequality itself. Both reviewer audits (issue #71, T1 round 2) independently verified this chain.

5. Robust-zero lower bounds (certified, restricted dictionary)

5.1 Fidelity-to-Bargmann bridge

Convention: for normalized $|\psi\rangle$, the Bargmann function $F_\psi(z) = e^{-|z|^2/2}$ -normalized coherent overlap satisfies $|F_\psi(z)| \leq e^{\{|z|^2/2\}}$, and the evaluation functional has norm $e^{\{|z|^2/2\}}$. If $F(\psi, \varphi) \geq 1-\varepsilon$, choose the

global phase of φ so $\langle \psi | \varphi \rangle \geq 0$; then $\|\psi - \varphi\|^2 = 2 - 2|\langle \psi | \varphi \rangle| \leq 2 - 2\sqrt{1 - \varepsilon} =: d(\varepsilon)^2$, and for all z

$$|F_\psi(z) - F_\varphi(z)| \leq e^{\{|z|^2/2\}} d(\varepsilon).$$

The phase change multiplies all dictionary coefficients by one phase and changes neither the component count nor the zeros.

5.2 Robust zeros

A **robust-zero certificate** at parameters (R, δ, ε) is a finite set of **distinct** zeros $z_1, \dots, z_J$ of F_ψ in $D(0, R)$, with multiplicities m_j , such that the closed disks $\bar{D}(z_j, \delta)$ are pairwise disjoint and each is **(δ, ε) -robust**:

$$\min_{\{|z - z_j| = \delta\}} |F_\psi(z)| > e^{\{(|z_j| + \delta)^2/2\}} d(\varepsilon).$$

Its value is $\sum_j m_j$. Theorem B' holds for **every** certificate, so the best exhibited certificate gives the best bound; we write **$N_{\text{robust}}(\psi; R, \delta, \varepsilon)$** for the value of the certificate actually exhibited (the census tables of §7.2 are such exhibits — N_{robust} is certificate-relative, not an intrinsic function of $(\psi, R, \delta, \varepsilon)$).

5.3 Theorem B'

Theorem B' (row D2). Let $|\psi\rangle$ be a normalized pure state and let $|\varphi\rangle$ satisfy $F(\psi, \varphi) \geq 1 - \varepsilon$ with $\chi_{\{G_{\text{eq}}(a, B)\}}(\varphi) \leq K$. Then

$$K \geq N_{\text{robust}}(\psi; R, \delta, \varepsilon) / C - B(R + \delta),$$

with C the absolute constant of Lemma B1'-eq. *Mixed version* (pure target): if σ is mixed with $F(\psi, \sigma) = \langle \psi | \sigma | \psi \rangle \geq 1 - \varepsilon$ and some generalized-ensemble decomposition has $\chi_{\{G_{\text{eq}}(a, B)\}}(\varphi_i) \leq K$ for μ -almost every i (equivalently the restricted roof is $\leq K$, which is attained by Lemma A1'), then the same bound holds.

Proof. By §5.1 the perturbation on each circle $|z - z_j| = \delta$ is strictly below $\min |F_\psi|$ there; Rouché gives F_φ the same number of zeros (with multiplicity) in $D(z_j, \delta)$ as F_ψ has, which is **at least** m_j (other zeros of F_ψ may also lie in the disk; no assumption excluding them is needed). The disks are disjoint and contained in $D(0, R + \delta)$, so F_φ has $\geq \sum m_j = N_{\text{robust}}$ zeros there; Lemma B1'-eq bounds that count by $C(K + B(R + \delta))$. Rearranged. Mixed case: since $\int |\langle \psi | \varphi_i \rangle|^2 d\mu \geq 1 - \varepsilon$, the set $\{i : |\langle \psi | \varphi_i \rangle|^2 \geq 1 - \varepsilon\}$ has positive measure — otherwise the average would be strictly below $1 - \varepsilon$ — and it meets the full-measure set where the rank condition holds; pick a member and apply the pure case. ■

5.4 Why the restriction is forced (impossibility remarks)

(a) No unrestricted disk-zero bound. The even-cat family $|\alpha\rangle + |-\alpha\rangle$ ($K = 2$, $a = 0$, $b = \pm\alpha$) has $\sim 2R\alpha/\pi$ zeros in $D(0, R)$. As $\alpha \rightarrow \infty$ the zero count in a fixed disk is unbounded at fixed K . Hence no function of (K, R) alone bounds disk zeros, so no lower bound on the unrestricted χ_G can be extracted from **raw disk zero counts alone**: the displacement budget B must appear. (Refinements that weight zeros by energy or position are not excluded by this family and might evade the obstruction; what is excluded is the count-only route. This is also why our certified quantity is $K_\varepsilon\{G_{\text{eq}}(a, B), F\}$; cf. §2.2.)

(b) The common-squeezing restriction and the plateau. With two distinct squeezings the family $e^{\{a z^2/2\}} - 1$ ($K = 2$) has $\sim |a|R^2/\pi$ zeros in $D(0, R)$ (a double zero at the origin — note Lemma P1 does not apply across distinct squeezings). The general two-component count — stated and calibrated **for centered disks only**, with the off-center obstruction and sharpness discussion — lives in one place, Appendix D; the body deliberately does not restate the formula. The consequence needed here: for *general* bounded-squeezing dictionaries the adversary budget acquires an area term $(K - 1) \cdot O(aR^2)$, and the lower bound for area-density targets saturates at $K \geq 1 + \Omega(1/a)$ instead of growing with R . We record this as a calibrated **conjecture** (statement in Appendix D), not a theorem; the certified results of this note are the common-squeezing case.

(c) Selectivity. For the cat target itself the bound is vacuous (its own zeros are line-density and absorbed by the $B(R + \delta)$ term) — correct, since cats *are* rank 2. The bound has content precisely for targets with **area-**

density robust zeros, the GKP class being the natural candidate (§7.2). The scaling limit is honest: as $a^* \rightarrow 1$ or $B \rightarrow \infty$ the technique loses all power.

6. Inequivalence of the two compressibilities (certified + empirical)

Target (**single mode**, like every certified and empirical row of this note — see the Modes column of §2.4): the one-mode thermal lossy cat of experiment 20, $\rho_T = N_\sigma(E_\eta(|\text{cat}\rangle\langle\text{cat}|))$, $\eta = 0.8$, $\sigma = 0.1$ (per-quadrature added variance), $\alpha = 1.5$, parity +1. The exp20 derivation proves Lemmas 1–2 and Theorems 1–2 in one mode; its closing section sketches a three-mode port but omits Theorem 2 from the porting list, so this note cites the one-mode statements only and makes no multimode claim.

Theorem C' (inequivalence; rows D1, D4, D5). (C'1, certified) $R_0(\rho_T) = \infty$ (with the $\min \emptyset = \infty$ convention of §2.3): for **no** $\eta' \in (0,1]$ does ρ_T admit a finite-rank pre-loss representation. [Experiment 20, derivation.md, Lemmas 1–2 and Theorems 1–2: regimes I/II by full-rankness of noise outputs, the boundary $\eta' = \eta - \sigma$ by an analytic-kernel linear-independence argument — the evaluation points there must avoid the cat Bargmann function's zero set, always possible since that set is discrete — regime III by the Bargmann-zero vs Q-positivity obstruction.] (C'2, certified) $\chi_G(\rho_T) \leq 2$, hence $K_{\varepsilon^{\{G,F\}}}(\rho_T) \leq 2$ for all ε : $|\text{cat}\rangle$ has $\chi_G \leq 2$ and both N_σ and E_η are Gaussian channels; apply Theorem A'. (C'3, empirical, row D5 — **not** an R_ε point) the rank-2 family fit approaches ρ_T to best-found cutoff-conditioned residuals $1-F = 3.44 \times 10^{-3} / 2.37 \times 10^{-3} / 2.17 \times 10^{-3}$ (Route B, $n_{\text{score}} = 12$, at $K = 2/4/8$ ket components per rank-2 column; the series over cutoffs is still increasing). These generalized fidelities are computed on cropped/subnormalized matrices and can exceed the true state fidelity, so no inequality on the R_ε of §2.3 is asserted; they are the empirical surrogate showing the boundary of C'1 is thin at finite resolution. Turning them into a genuine D4 point requires a full-state fidelity tail bound (open).

Reading (one-directional separation). One and the same state has Gaussian-superposition rank **at most** 2 and exact pre-loss operator rank $R_0 = \infty$ (the behaviour of R_ε for $\varepsilon > 0$ is not determined here): small χ_G does not imply small pre-loss operator rank. ($\chi_G(\rho_T) = 2$ exactly is not claimed — only the $\leq$ direction is proven; and whether small R forces small χ_G , the converse separation, is left open, so we do not claim “neither implies the other”.) Mechanically, the Gaussian classical noise cannot increase χ_G above the cat's value (Theorem A') while exploding operator rank — the continuous-ensemble roof absorbs what the finite-rank pre-image cannot. This recovers, in sharpened form, what experiment 20 actually proved — and corrects an earlier conflation of the two quantities in this repository's working notes (issue #71, C'3 correction).

7. Measured curves (empirical / proxy) and the GKP census (Gate E)

7.1 Experiment 22 artifacts (rows D5, D6)

From committed results only (no new fits): the **family-constrained approximation curve** (synthetic thermal-cat target; rank-2 $BB^\dagger$ + loss family, K ket components per column; all n_{score} series; labels and machine-readable `rank_primitive`, `fidelity_convention`, `cutoff_status`, `epistemic_status` in the JSONs) and the **real-data rank proxy** (held-out per-sample NLL vs operator mixture rank $R = 1..5$ on the public GKP homodyne dataset of Konno *et al.* [Science **383**, 289 (2024); data: Dryad doi:10.5061/dryad.t76hdr86j — ref 8']; true-state fidelity — hence K_ε — is inaccessible on measured data, and the proxy is labeled as such). The certified-lower-bound panel is reserved in the exp22 figure; the census values live in the exp23 artifacts (§7.2). Figures: experiments/22_kcurves/.

7.2 Gate E: robust-zero census for finite-energy GKP

For finite-energy GKP states (square lattice, envelope $\Delta \in \{0.2, 0.3, 0.4\}$, $\langle n \rangle \approx 12.0 / 5.1 / 2.4$ — reference-cutoff estimates, per the artifact) the zeros of the Bargmann function in energy windows $R = 0.75/1.00/1.25 \cdot \sqrt{\langle n \rangle}$ are located and tested against the robustness threshold of §5.2, with explicit truncation-tail bounds (experiments/23_gkp_robust_zeros). Two target rows, per the declaration table (neither is a full certification; see the numerical-status paragraph below):

- **D2 (ψ_{trunc}):** $N_{\text{robust}} = 12 / 8 / 4$ at the largest windows ($\varepsilon = 10^{-4}$, $\delta = 0.18$; at $\delta = 0.30$ the $\Delta = 0.2$ count drops to 8) — the count **increases with energy in the same direction across all three configurations** (the same triple, set against the $\langle n \rangle$ values quoted at the top of this subsection, gives ratios that are far from constant, and with R, Δ , tail and margins co-varying, three points cannot identify a proportionality law or a density constant — a $\Delta \rightarrow 0$ series with error estimates would be needed for an area-density *scaling* claim). All three points are nonzero. These counts instantiate the **premises** of the symbolic bound $K \geq N_{\text{robust}}/C - B(R+\delta)$ for the renormalized truncated states; no specific positive K value is certified, since C is deliberately not assigned a number and B is a free dictionary parameter.
- **D3 (untruncated comb, lifted threshold $d(\varepsilon) + \|\text{Fock tail}\| + 4 \cdot \|\text{lattice remainder}\|$ — the factor 4 converts the pre-normalization envelope amplitude to a normalized-state distance via $\|\psi\|/\|\psi\| - v/\|v\| \leq 2\|\psi-v\|/\|\psi\|$ with lattice-sum norm $\geq 1/2$ asserted at build; the term is $\sim 10^{-15}$ and no verdict depends on it, checked up to a 10^{12} -fold inflation):** all three configurations instantiate the **premises for the untruncated finite-energy comb** ($N_{\text{robust_lifted}} = 4 / 8 / 4$ at $\delta = 0.18$: $\Delta = 0.2$ at $\varepsilon = 10^{-4}$, $\Delta = 0.3$ and 0.4 at $\varepsilon \leq 10^{-3}$). An earlier draft reported that $\Delta = 0.2$ could not lift and misattributed this to a polynomial-degree ceiling; the binding constraint was in fact the Markov remainder of the Fock tail bound, controlled by an independent parameter (reference cutoff, now 3000; tails $3.3 \times 10^{-4} / 2.9 \times 10^{-5} / 6.2 \times 10^{-6}$), and the corrected exact tail distance $\sqrt{2-2\sqrt{1-p}}$ is used throughout. The degree ceiling $n_{\text{max}} \lesssim 300$ constrains only the zero search, not the tail.

Numerical status (the census is numerically supported, not certified). Two layers must not be conflated. (i) The circle-minimum step is *analytically discretized*: the reported lower bound is sampled minimum minus arc-step times a coefficient-norm Cauchy bound on $|F'|$ — but it is evaluated in float64 **without outward rounding**, so it is not a computer-verified proof; interval arithmetic would be needed for that. (ii) Zero *existence/count* inside each disk rests on polynomial root-finding with residual checks and tiled argument-principle cross-checks (interior cells) — numerics, and a Theorem B' premise. The JSON's `_bounded` columns therefore refer to layer (i) only, and the census as a whole is a **numerically supported instantiation of the theorem's premises**, not a conditional or full certification. Supporting evidence for the numbers themselves is strong: `bounded  $\leq$  sampled` holds across all rows with no robust zero lost (the historically tightest row, $\Delta = 0.4$ at $\varepsilon = 10^{-3}$, now carries lifting margins of 37.96% on the sampled minimum and 36.03% on the float64 discretized bound (no outward rounding — see above), the tightening of the tail bound having widened what was an 8.7%-class margin); independent verification (PR #113 review) found the derivative bound within a factor 1.7 of the true $\max|F'|$ and the 1440-point sampled minima unchanged to reported precision against 2×10^5 -point sampling on every zero — the discretized-bound step tightened the *status* of the same numbers, not the numbers. The lattice-envelope amplitude ($\sim 4 \times 10^{-16}$) is folded into the lifted threshold (entering pre-normalization; conservative in all three configurations, checked numerically).

Robustness activates only for $\varepsilon \lesssim 10^{-3}$ (at $\varepsilon = 10^{-2}$ no zero is robust in any configuration): the bound has content only under high-fidelity approximation demands. The enumerated coded claims are checked against the exp23 JSON by the claim checker (§7.3).

7.3 Claim checker (build gate)

The build script `docs/kepsilon-note/check_claims.py` verifies a **coded list** of the claims this note attributes to artifacts — including numeric claims asserted inside prose metadata fields such as `rank_primitive` and `lifting_rule` — against the committed JSONs of exp20/22/23. Coverage is the coded checks, not literally every number in the note (energies, some δ rows, and physical parameters are spot-checked only). The lesson behind it: three same-shaped failures (routeB's hardcoded verdict print, the K-axis conflation, the blanket ψ_{trunc} declaration) all came from prose carrying claims that no computation backed. Prose that makes a claim should get that claim coded into the checker — coverage is the coded list and grows with it, not a property of prose as such.

7.4 Quoting policy for Route B numbers

Cutoff series always quoted alongside (8/10/12); headline = largest scored cutoff; the phrase “cutoff-stable” is not used (the series still increases); the compressed range “ $1-2 \times 10^{-3}$ ” is deprecated repository-wide.

8. Discussion

The world-model question that motivates this repository — *is the world describable by Gaussian splats?* — here takes a sharpened, answerable form. The pure-state definitions are the community’s $(\chi, \chi_\delta, r^*_\varepsilon)$; the continuous ess-sup rank roof is our formulation of a question their extent-side analogues leave open. Within the literature we checked (Gate S survey plus the round-2/3 reviews’ primary sources — not a systematic novelty search), what this note adds is a certified way to say **no at a rate** for bounded dictionaries (area-density robust zeros defeat line-density zero budgets), an inequivalence showing that “how many Gaussians” and “what operator rank before loss” are different questions with opposite answers on the same state, and measured curves that keep the theory attached to data. The technique’s limits are part of the result: it dies at unbounded displacement, saturates for general squeezing (plateau conjecture), and is selective — vacuous on states that *are* Gaussian-compressible (the cat), though not only on those: with B a free dictionary parameter (§7.2), a large enough budget voids the bound for any target, so content always requires committing to a dictionary budget first.

A same-cluster preprint appeared during this note’s review: Wang et al. [arXiv:2607.06404] construct bosonic codes of *finite stellar rank*, GKP-adjacent but measuring a different resource — stellar rank of code states versus our equal-squeezing Gaussian-rank budget. The two quantities are defined on different decompositions and this note makes no claim about their relation (an implication either way would require an explicit state family, which we have not constructed); the census here is simply complementary to their construction.

Open next steps, in charter order: full certification of the census (interval arithmetic with outward rounding + validated winding-number integrals for zero existence, §7.2), the multimode extension (§2.2), the plateau conjecture, the dynamics program $K_\varepsilon(t)$ (issue #97 direction 2), and the relation between the zero-counting route and the low-rank-approximation route of [Cottier–Chabaud] — the two see different obstructions and might compose.

Appendix A. Proof of Lemma A1’

A.1 Setup. Fix M modes. Pure Gaussian states form a finite-dimensional real-analytic manifold $\mathcal{G}$ (parameterized by covariance/mean or, per mode pair, by (a, b) with $|a| < 1$ in the Bargmann form). For $K \in \mathbb{N}$ let

$$S_K := \{ |\psi\rangle \in \mathcal{H}, \|\psi\| = 1 : |\psi\rangle \propto \sum_{k=1}^K c_k |G_k\rangle, |G_k\rangle \in \mathcal{G} \}.$$

A.2 Measurability. The map $\Phi_K : \mathcal{G}^K \times \mathbb{C}^K \rightarrow \mathcal{H}$, $\Phi_K(G_1..G_K, c) = \sum c_k |G_k\rangle$, is continuous (Gaussian states depend continuously on parameters in norm; finite sums preserve continuity). S_K is the image of the Borel set $\{\|\Phi_K\| \neq 0\}$ under normalization composed with Φ_K , i.e. an analytic subset of the unit sphere of $\mathcal{H}$ (continuous image of a Polish space). Analytic sets are universally measurable, so $\{\psi : \chi_G(\psi) \leq K\} = S_K \pmod{\text{phase}}$ is universally measurable, and $\chi_G(\cdot) = \min\{K : \psi \in S_K\}$ (with $\min \emptyset = \infty$) is a universally measurable function into $\mathbb{N}_{\geq 1} \cup \{\infty\}$. For a generalized ensemble (standard Borel index space, measurable pure-state field) the composition $i \mapsto \chi_G(\psi_i)$ is measurable for the completed measure, and $\text{ess sup}_i \chi_G(\psi_i)$ is well-defined.

A.3 Attainment of the outer infimum. The roof objective takes values in the discrete set $\mathbb{N}_{\geq 1} \cup \{\infty\}$. (The same discreteness argument applies verbatim to the restricted roof $\chi_{G_{\text{eq}}(a,B)}$ of §2.2 — attainment there is not a separate assumption; measurability of the restricted rank sets follows as in A.2 with the atom manifold $\mathcal{A}(a,B)$ in place of $\mathcal{G}$.) Let $m := \inf$ over decompositions of the ess sup. If $m = \infty$ there is nothing to attain. If $m < \infty$, some decomposition has $\text{ess sup} < m + 1$, i.e. $\leq m$, i.e. $= m$ by definition of the infimum. (The ensemble set is nonempty: the spectral decomposition is a countable generalized ensemble.)

A.4 Monotonicity in ε and $K_0 = \chi_G$. The feasible set $\{\sigma : F(\rho, \sigma) \geq 1 - \varepsilon\}$ grows with ε , so the min is non-increasing. At $\varepsilon = 0$, $F(\rho, \sigma) = 1 \iff \rho = \sigma$ for density operators, so $K_0\{G, F\}(\rho) = \chi_G(\rho)$. ■

Appendix B. Re-derivation of the disk zero bound

Following [Friedland, arXiv:2606.24823, Lemma 3.2 and Lemma B.1]; both steps were independently audited in the T1 round-2 reviews (issue #71). The only external input is the classical interval Turán inequality:

Input (interval Turán). For a nonzero exponential sum $q(t) = \sum_{k=1}^m a_k e^{\mu_k t}$ restricted to a real interval I of length L , and a subinterval $J \subset I$,

$$\sup_I |q| \leq e^{\{L \cdot \max_k |\operatorname{Re} \mu_k|\}} (c L / |J|)^{m-1} \sup_J |q|,$$

with c an absolute constant. [Classical; also subsumed by the Turán–Nazarov inequality, Friedland–Yomdin arXiv:1107.0039, Thm 1.1.]

B.1 Disk growth (Friedland Lemma 3.2). For q as above, $z_0 \in \mathbb{C}$, $0 < r \leq R$, $\sigma := \max_k |\mu_k|$:

$$\sup_{D(z_0, R)} |q| \leq e^{\{2R\sigma\}} (C R/r)^{m-1} \sup_{D(z_0, r)} |q|.$$

Derivation. Let z^* realize the outer sup and w^* the point of $D(z_0, r)$ where $|q|$ is largest on the segment ray through z^* — parameterize the line through z_0 and z^* as $z = z_0 + s\theta$ with $|\theta| = 1$, $s \in [-R, R]$. On this line $Q(s) := q(z_0 + s\theta)$ is an exponential sum with exponents $\theta\mu_k$, and $|\operatorname{Re}(\theta\mu_k)| \leq |\mu_k| \leq \sigma$. Apply interval Turán with I of length $2R$ and J = the sub-segment of length $2r$ inside $D(z_0, r)$: sup on I (hence at z^*) is $\leq e^{\{2R\sigma\}} (cR/r)^{m-1}$ times the sup on J , which is $\leq$ the sup on $D(z_0, r)$. ■

B.2 Blaschke step (Friedland Lemma B.1). Suppose q has N zeros (with multiplicity) in $\bar{D}(z_0, r)$, $q \not\equiv 0$. Write $q = B_N \cdot h$ on $D(z_0, 8r)$ where B_N is the product of the N Möbius factors vanishing at those N zeros (normalized for the disk of radius $8r$) and h is holomorphic there (h may have further zeros of q outside $\bar{D}(z_0, r)$; they are not needed). Each Möbius factor has modulus $\leq \theta < 1$ (an absolute constant) on $\bar{D}(z_0, r)$ and $= 1$ on $\partial D(z_0, 8r)$, so

$$\sup_{D(z_0, r)} |q| \leq \theta^N \sup_{D(z_0, 8r)} |q| \leq \theta^N e^{\{16 r \sigma\}} (8C)^{m-1} \sup_{D(z_0, r)} |q|,$$

using B.1 at radii $(8r, r)$. Dividing by the (nonzero) sup and taking logarithms: $N \log(1/\theta) \leq 16 r \sigma + (m-1) \log(8C)$, i.e.

$$N \leq C' (m + \sigma r) \text{ with } C' \text{ absolute. } \blacksquare$$

Appendix C. Interconversion with χ_δ of HTFY on pure states

HTFY define $\chi_\delta(|\psi\rangle) = \inf\{\chi(|\psi'\rangle) : \|\psi - \psi'\| < \delta\}$ (vector-norm ball). For normalized ψ, ψ' with optimally aligned phases, $\|\psi - \psi'\|^2 = 2(1 - |\langle\psi|\psi'\rangle|)$ and $F = |\langle\psi|\psi'\rangle|^2$, so a fidelity ball $F \geq 1 - \varepsilon$ equals a norm ball of radius $d(\varepsilon) = \sqrt{2 - 2\sqrt{1 - \varepsilon}}$:

$$\chi_{\{d'\}}(|\psi\rangle) \leq K_\varepsilon \{G, F\}(|\psi\rangle) \text{ restricted to pure approximants} \leq \chi_{\{d(\varepsilon)\}}(|\psi\rangle) \text{ for every } d' > d(\varepsilon)$$

(the sandwich, rather than an equality, because HTFY's ball is open and ours closed: an open ball of radius $d(\varepsilon)$ is contained in our closed ball, so the min over it is *larger*, $\chi_{\{d(\varepsilon)\}} \geq K_\varepsilon$; and our closed ball sits inside any open ball of radius $d' > d(\varepsilon)$, giving the left inequality. The two sides agree except at the at-most-countably-many radii where χ_δ jumps). Two caveats keep this from being an identity of the full quantities: (i) our K_ε permits **mixed** approximants σ , which can only lower the min (on pure targets, if a mixed σ with roof value K satisfies $\langle\psi|\sigma|\psi\rangle \geq 1 - \varepsilon$, the averaging argument of Theorem B' produces a pure member with the same K and fidelity, so the two agree on pure targets); (ii) HTFY's χ counts superpositions only, matching our pure-state χ_G . On mixed targets the quantities differ by construction (roof vs no natural χ_δ analogue), and no identification is claimed.

Appendix D. The plateau conjecture for general bounded-squeezing dictionaries

Calibrating examples (certified). (i) $e^{\{a z^2/2\}} - 1$ (two components with squeezing gap a) has $\approx |a|R^2/\pi$ zeros in $D(0, R)$, including a double zero at 0. (ii) More generally, for two components $c_1 e^{\{a_1 z^2/2 + b_1 z + d_1\}} + c_2 e^{\{a_2 z^2/2 + b_2 z + d_2\}}$ with parameter gaps $\Delta x := x_2 - x_1$ (assuming $f \neq 0$, i.e. excluding identical parameters with $c_2 = -c_1$), zeros solve $\Delta a \cdot z^2/2 + \Delta b \cdot z + \Delta d \in \log(-c_1/c_2) + 2\pi i\mathbb{Z}$ — a lattice of spacing 2π on a line, shifted

by the coefficient ratio; counting lattice points hit by a degree-2 polynomial image of the **centered** disk $\bar{D}(0, R)$ gives $N \leq C(1 + |\Delta a| R^2 + |\Delta b| R)$, sharp in form (the additive constant is again necessary). **Off-center disks are worse:** over $\bar{D}(z_0, R)$ the image diameter grows like $|\Delta a| |z_0| R$, so the count acquires an unavoidable $a/z_0 R$ term (verified numerically: for $e^{\{0.5 \cdot z^2/2\}} - 1$ the unit disk at $|z_0| = 10^3$ contains ≈ 159 zeros, matching the ray density $a/z_0/(2\pi)$). The common-squeezing slice is immune — there the Gaussian factor divides out globally and the budget is center-uniform. Any general-dictionary zero bound must therefore carry an area term $a R^2$.

Conjecture D. For $f = \sum_{k=1}^K c_k e^{\{a_k z^2/2 + b_k z + d_k\}}$ with $|a_k| \leq a^* < 1$, $|b_k| \leq B$, $f \not\equiv 0$, in minimal representation, on the **centered** disk:

$$N(\bar{D}(0, R); f) \leq C(K-1)(1 + a^* R^2 + B R).$$

(For off-center disks the right side must additionally carry a $C(K-1) \cdot a^* \cdot |z_0| \cdot R$ term — see the calibrating example above; a center-uniform bound without that term is false.)

Consequence if true. For an area-density target ($N_{\text{robust}} \sim \rho_0 R^2$, $\rho_0 \leq 1/\pi$) the Theorem-B'-style bound saturates: $K \geq 1 + \rho_0/(C a^*)$ as $R \rightarrow \infty$ — a plateau inversely proportional to the squeezing budget, instead of unbounded growth. The certified results of this note are the a-common slice, where the area term cancels exactly (§4); the conjecture marks where the technique's power ends, not where the states become easy.

Status. Statement calibrated by the examples above (T1, issue #71); proof route sketched via the order- K linear ODE with polynomial coefficients satisfied by f (Wronskian $= (\prod_k f_k) \cdot Q$ with $\deg Q \leq K(K-1)/2$) and disconjugacy on disks avoiding Q 's zeros; the naive complex Chebyshev shortcut is false ($W[1, e^{\{\mu z\}}]$ never vanishes while $e^{\{\mu z\}} - 1$ has $\sim R\mu/\pi$ disk zeros), so coefficient bounds must enter. Left open.

References

Primary-source verification status per repository policy: [pdf] = read in PDF/HTML by this project; [stmt] = the specific cited statement was transcribed and checked; remaining items to re-verify at final draft.

1. K. Hahn, R. Takagi, G. Ferrini, H. Yamasaki, *Classical simulation and quantum resource theory of non-Gaussian optics*, arXiv:2404.07115; Quantum 9, 1881 (2025). [pdf][stmt: χ , χ_δ , ξ definitions; convex-roof mixed extension; monotonicity under Gaussian unitaries/measurements]
2. K. Hahn, A. Garnier, G. Ferrini, A. Ferraro, U. Chabaud, *Assessing non-Gaussian quantum state conversion with the stellar rank*, arXiv:2410.23721; Quantum 10, 2095 (2026). [pdf][stmt: Def. 7 r^*_{ϵ} ; Thm 1 monotonicity; convex-roof mixed stellar rank]
3. F. Cottier, U. Chabaud, *Lower Bounds on Coherent State Rank*, arXiv:2604.00766 (2026). [pdf][stmt: ϵ -approximate coherent state rank; low-rank-approximation and permanent-complexity lower bounds; single-mode characterization — full text confirmed at round 2 review (coherent-only dictionary; technique disjoint from the zero route)]
4. O. Friedland, *A Disk-Growth Remez Principle and a Modular Proof of the Measurable Turán-Nazarov Inequality*, arXiv:2606.24823 (2026). [stmt: Lemma 3.2, Lemma B.1 — audited by two independent reviews; re-derived in Appendix B]
5. O. Friedland, Y. Yomdin, *An observation on Turán-Nazarov inequality*, arXiv:1107.0039. [pdf][stmt: Thm 1.1 interval Turán-Nazarov]
6. M. Motamedi et al., *The stellar decomposition of Gaussian quantum states*, arXiv:2504.10455. [stmt: Bargmann parameter table, cross-check of $a = -e^{\{i\varphi\}} \tanh r$, $b = \alpha - a\alpha^*$] 6'. Y. Wang, A. Udupa, T. Hillmann, U. Chabaud, A. Ferraro, G. Ferrini, *Bosonic quantum error-correcting codes with finite stellar rank*, arXiv:2607.06404 (2026). [flagged at round-2 review: GKP with finite stellar rank from the same cluster; contrasted in §7.2/§8 — their object is stellar rank of code states, ours the equal-squeezing Gaussian rank budget; neither implies the other]
7. Experiment 20 derivation note (this repository, experiments/20_noninclusion/derivation.md): Lemmas 1–2, Theorems 1–2. [in-repo, PR-64-reviewed]
8. Experiments 22–23 artifacts (this repository): exp22 curves JSON, exp23 robust-zero census JSON (`_sampled/_bounded` columns). [in-repo, reviewed] 8'. S. Konno et al., “Logical states for fault-tolerant quantum computation with propagating light,” *Science* **383**, 289 (2024). Data: Dryad,

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